

GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

QIAN QI

ABSTRACT. A finite observer of an uncertain causal experiment must use the same update rule at every time and at every inaccessible model parameter. We prove a discounted infinite-horizon transfer theorem for this common controller, starting from the report and payoff laws of finite interventions. Finite-history product domination makes the response of every fixed-cardinality controller continuous in a compact topology, even with infinite-dimensional hidden dynamics and changing report supports. Consequently the robust optimum is attained, finite model families determine its value, and finite report tables give two-sided certificates at the unchanged persistent-state budget under effective response data. No finite-dimensional predictive carrier, filter contraction, or equivalent active report supports is required for this continuing interface. Nonlinear hidden-state kernels and noncommuting projective instruments verify the theorem independently. A quantitative consequence gives sharp discount–expansion moduli for unstable hidden dynamics. Singular acquired laws yield exact blind-memory curves, including a nondominated preparation class. A continuous-report example supplies a complete rational certificate. Calibration, numerical error, causal deficiency, persistent labels, program description, workspace, acquisitions and physical time are accounted for separately. Discounted continuation is not an exact internal deadline or an undiscounted average-cost assertion.

1. THE CAUSAL EXPERIMENT AND ITS MATHEMATICAL QUESTION

An experiment does not supply a manifold to an observer. It supplies a preparation, interventions, reports, and executable future tests. Predictive equivalence is determined by those tests; predictive geometry is the law actually acquired through a legal execution. The question is which part of that law one finite observer can retain, and how a resource-certified simulation changes the answer. Neither the dimension of a formal parameter space nor a separate optimum at each parameter answers this question.

There are two distinct difficulties. First, repeated use of one report-to-state rule couples all times. Secondly, the model parameter is fixed throughout the experiment but inaccessible to the controller. A sequence of finite-horizon or parameterwise minimizers need not be a single reusable rule. The center of this paper is a compactification which respects both constraints. Its input is the raw response of the experiment to fixed finite intervention words, not a chosen finite-dimensional posterior coordinate system.

The main theorem, Theorem 2.2, establishes an attained robust value for every fixed finite alphabet and every strictly discounted infinite horizon. Proposition 2.3 is its mechanism: a report row is reused at different report coordinates, so finite-history product domination permits tensor testing, even though pointwise multiplication is not weak-star continuous. Theorem 3.3 then gives an implementing finite table and a lower certificate against all Borel controllers of the same size. This is stronger than evaluating a selected controller list, but it is not an efficient general synthesis algorithm.

Date: 9 October 2026.

2020 Mathematics Subject Classification. Primary 62B05, 93E20; Secondary 60G35, 94A34, 81P45.

Key words and phrases. causal experiment, predictive quotient, finite-state controller, discounted minimax risk, acquired geometry, changing support, quantum instrument.

The interface matters. Here all primitive actions are legal at every continuing call, and the controller does not have to discover an exact internal deadline. Therefore small report errors may create previously impossible report paths without making the controller illegal. This is the precise reason that equivalent active-cell supports, needed for exact interface preservation in the finite-horizon theory, are unnecessary here. Finite-history product domination remains a substantive assumption. We do not infer domination of infinite path laws, average-cost convergence, or arbitrary stopping-time guarantees.

1.1. Hypothesis and conclusion hierarchy. The following diagram is part of the mathematical statement, not an identification of its different levels. In each row the same controller is selected before the model supremum.

Level	Additional data	Conclusion
Raw exact, Theorem 2.2	Finite actions and labels; finite-history product domination; compact continuous readouts; $0 < \beta < 1$	Attainment; finite-model value identity; uniform prefix approximation
Effective, Theorem 3.3	Common computable partitions; uniform response errors; model and readout nets; certified integrations	Same- M finite table and globally valid lower bound
Geometric, Theorem 5.1	A blind-storage interface and actual preparation laws; covering and mass certificates for rates	Exact robust quantization; matching singular memory laws
Transfer, Theorem 6.1	A typed simulator for scored prefixes; state and acquisition certificates	Discount-weighted deficiency and resource composition

There is no finite-dimensional physical hypothesis in the first row. No common-support equivalence is imposed in the second row; continuing legality is imposed instead. The third row is a quantitative realization principle, not a premise of the first two rows.

1.2. Position relative to earlier results. Static reduction and compactness of strategic measures are established tools [6, 9, 8, 3]. Robust finite-state control is also an established subject [2, 1]. Our claim is not that common controllers, compactness, or finite enumeration are new individually. The result combines a fixed reused alphabet, an arbitrary fixed-model family, continuous changing-support reports, and physical presentations not assumed finite-dimensional, with a global same-budget effective comparison. The common row is never replaced by independent rows at successive times.

The finite-horizon allocation, exact-deadline and geometric transfer results are preserved in the complete companion [5]. They are not unproved premises of the new long-run theorem. In particular, the exact-deadline expression involving $M - n$ is not applied to a continuing M -state controller. The companion contains the earlier predictive-realization and acquired-mass hypotheses in full. The next section also states exactly which predictive construction is used here.

2. RAW RESPONSES AND A COMMON INFINITE-HORIZON CONTROLLER

2.1. Prepared experiments, tests and acquired laws. Let $\Theta \neq \emptyset$ be a set. Its realized element is fixed for an entire execution. At time t , a standard Borel hidden state, a finite action $a \in A$, and a positive normalized causal kernel produce the next hidden state and a report in a standard Borel space Y_a . Preparation is prescribed at every θ . Hidden state may include the necessary physical history and may be infinite-dimensional. All actions in A are legal at every call. Positive here means nonnegative, not strictly positive. Every call is one raw acquisition; heterogeneous physical times may be included in the report with a separate cost certificate.

At time $t \geq 1$ a scored outcome Z_t is available to the declared task but is not additional feedback to the controller. A decision $d \in D$ incurs $0 \leq \ell(d, Z_t) \leq 1$. The space D is compact metrizable

and $d \mapsto \ell(d, z)$ is continuous for every z . A score depending on earlier physical history or on the action word is allowed. Its law, like the report law, is fixed by the raw experiment. No score may depend on a private controller seed except through the declared decision. Terminal lotteries can be included in D only with their sampling cost counted.

An executable future test consists of a legal finite continuation and a bounded outcome. A countable determining test class gives a Borel evaluation map

$$h \mapsto q(h) = (\mathbb{E}[f_r \mid h])_{r \geq 1} \in [0, 1]^{\mathbb{N}},$$

after fixing a preparation or a Bayes preparation of the parameter. The countable class must determine the desired tests and include the legal interface. Its evaluation fibers are precisely predictive equivalence for that class. If another statistic is sufficient in the sense that these conditional expectations are measurable functions of it, coordinatewise factorization gives the evaluation map as a measurable function of that statistic. This is the minimality assertion used here. It is not a claim that every quotient image is Borel or admits a compact autonomous update. When those additional realization properties are needed, the compact executable-test closure hypotheses in [5] apply.

Under an actually deployed controller γ , write $\nu_{\theta,t}^\gamma = \text{Law}_\theta^\gamma(q(H_t))$ whenever this evaluation is fixed, and

$$(1) \quad \bar{\nu}_{\theta,\beta}^\gamma = (1 - \beta) \sum_{t \geq 1} \beta^{t-1} \nu_{\theta,t}^\gamma.$$

These are probability laws of real preparations and reports. Conditional predictive states under a model are analyst variables, not parameter registers. Pooling by a retained label is conditional expectation under this very execution. Different controllers can acquire different laws. For the robust theorem below we compare total task risks, not a sum of separately optimized modelwise Bayes excesses.

2.2. Finite-intervention responses. For $\mathbf{a} = (a_1, \dots, a_t)$ let $P_{\theta,t}^{\mathbf{a}}$ be the law of the reports when this action word is imposed without feedback. Define the positive payoff measure

$$(2) \quad \Lambda_{\theta,t}^{\mathbf{a},d}(B) = \mathbb{E}_\theta^{\text{do}(\mathbf{a})}[1_{\{Y_{1:t} \in B\}} \ell(d, Z_t)] \leq P_{\theta,t}^{\mathbf{a}}(B).$$

Assume that for each a one probability measure σ_a satisfies

$$(3) \quad P_{\theta,t}^{\mathbf{a}} \ll \sigma_{\mathbf{a}} := \sigma_{a_1} \otimes \dots \otimes \sigma_{a_t} \quad \text{for every } \theta, t, \mathbf{a}.$$

The densities $f_{\theta,t}^{\mathbf{a},d} = d\Lambda_{\theta,t}^{\mathbf{a},d}/d\sigma_{\mathbf{a}}$ are in L^1 . Moreover

$$(4) \quad d_j \rightarrow d \implies \|f_{\theta,t}^{\mathbf{a},d_j} - f_{\theta,t}^{\mathbf{a},d}\|_1 \rightarrow 0.$$

Indeed the left side is bounded by the expectation of $|\ell(d_j, Z_t) - \ell(d, Z_t)|$, which tends to zero by dominated convergence. If the scored laws are specified directly rather than by a common Z_t , they are required to be positive submeasures of the same report law and (4) is an assumption. This permits an executable decision-dependent audit at call t . Only that audit is performed in a scored run, which then ends; incompatible earlier audits are not assumed to have been jointly performed. The continuing unscored kernels determine the report history before this audit. The parameter set need not be compact or itself a measurable space for the exact assertion; joint domination is common to all its elements.

Definition 2.1 (Reused Borel controller). An M -state controller, $M \geq 1$, has initial state $I_0 = 1$, action rows $\alpha_i \in \Delta(A)$, Borel rows $k_{ia} : Y_a \rightarrow \Delta(\{1, \dots, M\})$, and readouts $d_i \in D$. At a call it samples a_t from $\alpha_{I_{t-1}}$, observes Y_t , samples I_t from $k_{I_{t-1}, a_t}(Y_t)$, and issues d_{I_t} . Fresh sampling is conditionally independent given the specified rows. All these rows and readouts are identical across time and parameters. A remembered seed, counter, or other register must be part of I_t . No

external phase index is an argument of a row. The exact layer permits noncomputable Borel rows; a finite program is asserted only in the effective layer.

Let Γ_M be this class and fix $0 < \beta < 1$. Put

$$(5) \quad J_{\theta,\beta}(\gamma) = (1 - \beta) \sum_{t \geq 1} \beta^{t-1} \mathbb{E}_\theta^\gamma \ell(d_{I_t}, Z_t),$$

$$(6) \quad V_\beta^\Theta(M) = \inf_{\gamma \in \Gamma_M} \sup_{\theta \in \Theta} J_{\theta,\beta}(\gamma),$$

$$V_{\beta,N}^\Theta(M) = \inf_{\gamma \in \Gamma_M} \sup_{\theta \in \Theta} (1 - \beta) \sum_{t=1}^N \beta^{t-1} \mathbb{E}_\theta^\gamma \ell(d_{I_t}, Z_t).$$

The finite-prefix infimum is over the same continuing rows, not over a larger class of phase-dependent finite-horizon policies.

Theorem 2.2 (Discounted raw-response transfer). *Under (2)–(4), for every finite M and every $0 < \beta < 1$, the infimum $V_\beta^\Theta(M)$ is attained by one Borel controller. It is unchanged by any replacement of the physical realization preserving all raw report and payoff responses. Furthermore*

$$(7) \quad V_\beta^\Theta(M) = \sup_{\emptyset \neq F \subset \Theta, |F| < \infty} V_\beta^F(M),$$

$$(8) \quad V_{\beta,N}^\Theta(M) \leq V_\beta^\Theta(M) \leq V_{\beta,N}^\Theta(M) + \beta^N.$$

Thus every strict risk target below the robust optimum is excluded by finitely many fixed models. No witness-cardinality bound follows from this assertion alone. No finite-dimensional physical or predictive space, report-support equivalence, or filter stability is assumed. All M states can be reused indefinitely; there is no uncounded controller clock.

Proposition 2.3 (Tensor continuity of raw responses). *Equip the simplex-valued rows with the weak-star topology of $L^\infty(\sigma_a)$, action rows with their Euclidean topology, and D^M with its compact topology. This makes Γ_M compact, and $\gamma \mapsto \mathbb{E}_\theta^\gamma \ell(d_{I_t}, Z_t)$ continuous for each fixed θ, t .*

Proof. The sets $k_j \geq 0$, $\sum_j k_j = 1$ are weak-star closed in the product of unit balls, hence compact. Every equivalence class has a Borel version on a standard Borel probability space. Complete the finitely many rows on the union of their reference-null exceptional sets by the fixed point mass at label 1. Product domination makes this one completion harmless for all parameters and finite histories.

We use the following elementary tensor fact, for sequences or nets. If $u_{s,\lambda} \rightarrow u_s$ weak-star in $L^\infty(\sigma_s)$ and all have absolute value at most one, then for each $g \in L^1(\bigotimes_{s=1}^t \sigma_s)$,

$$\int g(y) \prod_{s=1}^t u_{s,\lambda}(y_s) d\sigma(y) \longrightarrow \int g(y) \prod_{s=1}^t u_s(y_s) d\sigma(y).$$

For a product of indicator functions this is a product of converging one-coordinate integrals. Finite sums of rectangle indicators are dense in L^1 of the finite product. The error in replacing g by such a sum is at most its L^1 error, uniformly in the net. This proves the fact, also when several factors are copies of the same changing row.

Write $i_0 = 1$ and expand the raw experiment along its finite action and state words. Causal iteration gives exactly

$$(9) \quad \mathbb{E}_\theta^\gamma \ell(d_{I_t}, Z_t) = \sum_{\mathbf{a}, i_1, \dots, i_t} \left(\prod_{s=1}^t \alpha_{i_{s-1}}(a_s) \right) \int f_{\theta,t}^{\mathbf{a}, d_{i_t}}(y) \prod_{s=1}^t k_{i_{s-1}, a_s}(i_s | y_s) d\sigma_{\mathbf{a}}(y).$$

This is a raw interventional expansion of the actual adaptive law; it does not run the adaptive controller on an unrelated full-history law. Tensor continuity treats each fixed readout. When

readouts also vary, (4) bounds the extra integral by an L^1 difference. There are finitely many terms at fixed t . Continuity follows. Repeated rows are tested at distinct product coordinates, not multiplied on a single reference coordinate. $\square$

Proof of Theorem 2.2. The tails of the discounted series are at most β^N , uniformly in every parameter and controller. Each fixed-model discounted risk is therefore continuous on the compact controller space by Proposition 2.3. Its supremum over an arbitrary parameter set is lower semicontinuous and attains its minimum. The expansion proves invariance under equal responses.

For $r \in \mathbb{R}$, let $C_\theta(r) = \{\gamma : J_{\theta,\beta}(\gamma) \leq r\}$. These are closed subsets of one compact space. If all finite model subfamilies have value at most r , their finite intersections are nonempty. The finite-intersection property gives a controller in the full intersection. Apply this to any r strictly greater than the supremum on the right of (7); the reverse inequality is inclusion of model sets. A strict target below the full value must fail on some finite subfamily. This argument concerns one shared variable; it is not a minimax interchange. Finally the pointwise bound between the full and truncated sums proves (8) by taking suprema and infima in their original order. $\square$

Remark 2.4 (Why the domination hypothesis is genuinely temporal). Uniform one-time marginals do not imply (3). If $Y_1 = Y_2$ is uniform on $[0, 1]$, both marginals have densities, but the pair is supported on the diagonal. For the alternating dyadic indicators u_m one has $u_m \rightarrow^* 1/2$, whereas $\int u_m^2 = 1/2 \neq 1/4$. Our proof would be invalid for this pair. Conversely, finite-product domination need not imply domination of the whole infinite path law. The theorem uses only finite products followed by a uniform discounted tail.

3. TWO-SIDED FINITE-RESOURCE IMPLEMENTATION

The exact theorem concerns Borel controllers. This section imposes additional effective data and constructs finite tables. Its comparison class remains every controller in Definition 2.1 at the same M .

3.1. Independent error certificates. Fix a prefix length N . For each action let $\mathcal{P}_{a,h}$ be a finite Borel partition, with decidable cell membership and computable reference masses. Reference-null cells may be assigned any row. Let $C_{\mathbf{a},h}$ be conditional expectation on the product of these partitions under $\sigma_{\mathbf{a}}$. Suppose supplied bounds satisfy

$$(10) \quad \sup_{\theta,d} \|f_{\theta,t}^{\mathbf{a},d} - C_{\mathbf{a},h} f_{\theta,t}^{\mathbf{a},d}\|_1 \leq \eta_{t,\mathbf{a},h}, \quad 1 \leq t \leq N,$$

and tend effectively to zero for each finite prefix. Define

$$(11) \quad E_N(h) = (1 - \beta) \sum_{t=1}^N \beta^{t-1} \min \left\{ 1, M \sum_{\mathbf{a} \in A^t} \eta_{t,\mathbf{a},h} \right\}.$$

The possibly large action-word factor is explicit. No efficient estimate is implied merely by existence of the tensors.

For the effective layer only, Θ is a computably compact metric family with finite nets $\Theta_s \subset \Theta$. We require a supplied controller-uniform bound

$$(12) \quad |J_{\theta,\beta}^N(\gamma) - J_{\theta',\beta}^N(\gamma)| \leq \Omega_N(s) \quad (d(\theta, \theta') \leq s)$$

for every Borel controller, with $\Omega_N(s) \rightarrow 0$ effectively. One sufficient certificate replaces $\eta_{t,\mathbf{a},h}$ in (11) by a uniform L^1 difference of the two response densities. Raw-kernel couplings often give much smaller bounds, as in Section 4.

Let $D_\rho \subset D$ be a finite loss net: each d has $d' \in D_\rho$ with $\sup_z |\ell(d, z) - \ell(d', z)| \leq \omega_D(\rho)$. Require certified finite-prefix risk integration for every listed table and model. These are extra computational assumptions, not consequences of compactness alone.

Lemma 3.1 (One common cell average). *For every $\gamma \in \Gamma_M$, average each row k_{ia} once on each σ_a cell, obtaining $\bar{\gamma}$ with the same action rows and readouts. Then*

$$\sup_{\theta} |J_{\theta, \beta}^N(\gamma) - J_{\theta, \beta}^N(\bar{\gamma})| \leq E_N(h).$$

Every averaged row is a legal continuing controller for every model, without equivalent active-cell supports.

Proof. For a fixed action and label word, conditioning on a product cell gives

$$\int (C_{\mathbf{a}, h} f) \prod_s k_s d\sigma_{\mathbf{a}} = \int f \prod_s \bar{k}_s d\sigma_{\mathbf{a}}.$$

All repeated occurrences of a row use the same one-coordinate cell average. In the difference of (9), group label paths by their final label j . At a fixed report word their nonnegative controller weights, summed over all label paths for a fixed action word, are at most one. Consequently the absolute difference is bounded by

$$\sum_{\mathbf{a}} \sum_{j=1}^M \|f_{\theta, t}^{\mathbf{a}, d_j} - C_{\mathbf{a}, h} f_{\theta, t}^{\mathbf{a}, d_j}\|_1 \leq M \sum_{\mathbf{a}} \eta_{t, \mathbf{a}, h}.$$

Clipping at one and summing the discount weights proves the assertion. All actions are universally legal and every row is stochastic. There is neither a forbidden newly created path nor an exact stopping constraint to violate. No equivalence of conditional null sets is used. $\square$

Lemma 3.2 (Finite probability tables). *Let $Q = 2^p \geq 1$. Every cell controller can be approximated by rows with entries in $Q^{-1}\mathbb{Z}$ and readouts in D_ρ , at the same M , with uniform prefix risk error at most*

$$(13) \quad B_N(Q, \rho) = (1 - \beta) \sum_{t=1}^N \beta^{t-1} \min\{1, t(|A| + M)/Q\} + \omega_D(\rho)(1 - \beta^N).$$

In particular $B_N \leq (|A| + M)/(Q(1 - \beta)) + \omega_D(\rho)$.

Proof. Round a simplex row by flooring Q times each entry and allocating the remaining integer units among the largest fractional parts, with deterministic tie-breaking. An r -entry row changes in total variation by at most r/Q . Couple the action and next-label samples until their first discrepancy, using the same physical kernels while the samples agree. Up to call t the probability of a discrepancy is at most $t(|A| + M)/Q$. Outside that event, only the specified readout error remains. Bounded loss and discount summation give (13). No support condition is necessary for legality in this continuing class. $\square$

Theorem 3.3 (Same-budget certified implementation). *Assume the effective data above. Let $B_a = |\mathcal{P}_{a, h}|$, including all positive reference cells. Enumerate all action rows and report rows of denominator Q and readouts in D_ρ . The complete list $\mathcal{G}$ has at most*

$$(14) \quad |D_\rho|^M \binom{Q + |A| - 1}{|A| - 1}^M \prod_{a \in A} \binom{Q + M - 1}{M - 1}^{MB_a}$$

members. For each $g \in \mathcal{G}$ and $\theta_i \in \Theta_s$, compute $l_{g, i} \leq J_{\theta_i, \beta}^N(g) \leq u_{g, i}$ with $u_{g, i} - l_{g, i} \leq \zeta$. Set $L = \min_g \max_i l_{g, i}$ and $U = \min_g \max_i u_{g, i}$. Then

$$(15) \quad \boxed{\max\{0, L - E_N - B_N\} \leq V_\beta^\Theta(M) \leq \min\{1, U + \Omega_N(s) + \beta^N\}, \quad U - L \leq \zeta.}$$

A minimizer of $\max_i u_{g, i}$ is one implementing table on the original family. Its state alphabet is still M ; it need not know θ or count to N . The total certified optimality gap is at most $E_N + B_N + \Omega_N(s) + \beta^N + \zeta$. The endpoints converge effectively by first choosing N and then

refining the supplied data. This is a global lower bound against all Borel controllers, not a local search certificate.

Proof. Let R_N be the Borel prefix value, and $R_{N,G}^\Theta$ its restriction to the complete list. Lemmas 3.1 and 3.2, uniformly over θ , give

$$R_N \leq R_{N,G}^\Theta \leq R_N + E_N + B_N.$$

Let $S_G = \min_g \max_{\theta_i \in \Theta_s} J_{\theta_i, \beta}^N(g)$. Net inclusion and (12) give $S_G \leq R_{N,G}^\Theta \leq S_G + \Omega_N(s)$ and $L \leq S_G \leq U$. A minimizer of the lower envelopes has every upper envelope at most its lower envelope plus ζ , so $U - L \leq \zeta$. The selected upper minimizer has full risk at most $U + \Omega_N(s) + \beta^N$. Combining these inequalities with (8) proves (15). The model modulus is needed on the upper endpoint only. Counting integer simplex rows gives (14). All integrations use one fixed model for the whole prefix. The model net and the enumeration are offline certificates, not information delivered to the running observer. $\square$

Remark 3.4 (Why effective response data can exist). For a finite prefix, if $(\theta, d) \mapsto f_{\theta, t}^{\mathbf{a}, d}$ has compact image in L^1 and the partitions generate the report sigma fields, their conditional expectations converge uniformly on that compact image. To see this, cover the image by finitely many L^1 balls; use convergence at the centers and the L^1 contraction of conditional expectation. Effectivity still needs computable moduli and integrations. This observation permits vanishing and changing densities. It does not turn bare domination into an algorithm.

3.2. Representation costs. Put $P = M|A| + M^2 \sum_a B_a$. A direct integer-table representation uses at most

$$P \lceil \log_2(Q + 1) \rceil + M \lceil \log_2 |D_\rho| \rceil$$

bits, plus descriptions of partitions, readouts and interpreter. The live cut state uses $\lceil \log_2 M \rceil$ bits. With a read-only random-access table, row sampling and lookup require $O(p + \log M + \log |A| + \log(1 + \max_a B_a))$ temporary bits in addition to the supplied classification and readout workspace. Sampling scans at most $|A| + M$ entries per call; sequential-tape access must pay its actual scan cost. Fresh p -bit uniforms implement dyadic rows exactly. Report parsing, comparison near cell boundaries, and physical acquisition times are separate supplied costs; exact access to an arbitrary real report is not asserted to be finite-bit hardware. Numerical classification error is handled in Section 6. The list size times the model-net size bounds the number of offline evaluations, not the time of each one. None of these upper accounts is claimed to be a matched complexity lower bound.

4. TWO RAW REALIZATIONS AND A QUANTITATIVE NONLINEAR CONSEQUENCE

4.1. Nonlinear hidden-state observations.

Theorem 4.1 (Arbitrary hidden dynamics with dominated observations). *Let X_t be standard Borel, with arbitrary prepared causal hidden-state transitions, and suppose each conditional report law has a density with respect to the same σ_a , at every hidden history, time and model. Assume the bounded readout loss is continuous in its compact decision variable. Then Theorem 2.2 applies, without compactness of X_t , a finite-dimensional filter, or contraction. Uniform effective finite-prefix response certificates imply Theorem 3.3.*

Proof. Under a fixed action word, the conditional report law given the observed past is a mixture of the dominated report kernels, and is therefore dominated by σ_{a_t} . Iterated disintegration on standard Borel spaces gives conditional Radon–Nikodym densities; multiplying them gives a density of the finite report word with respect to $\sigma_{\mathbf{a}}$. Repeated normalization makes its integral one. This induction also permits hidden transitions depending on past reports; it does not condition

on a hidden path as though that path were independent of the reports. The payoff measure is dominated by the report law, so it has an L^1 density as well. Dominated convergence gives (4). The raw response hypotheses, rather than any normalized posterior stability, now give the assertions. $\square$

This covers additive nonsingular Gaussian observations of a nonlinear system, with a positive reference density on the full report space. It also covers interval-noise observations whose support moves with the hidden state and parameter. Gaussian tails remain real report events; they are not removed by conditioning. The hidden state can be a function-valued variable in a separable Banach space.

The next result supplies quantitative information beyond effective approximability. Suppose a family on a common bounded metric hidden space has one measurable causal coupling of its raw innovations, valid for every commonly and adaptively selected action, so that until reports first differ

$$(16) \quad d_X(X_t^\theta, X_t^{\theta'}) \leq C_0 L^t s, \quad L > 1,$$

when the parameters and preparations are s apart. Suppose the one-call reports can be coupled with mismatch probability at most $C_1(d_X + s)$. On matching reports require the conditional expectation of $\sup_{d \in D} |\ell(d, Z_t) - \ell(d, Z'_t)|$ to be at most $C_2(d_X + s)$, conditional on the paired past, states and matched reports. In particular this holds for a hidden target with a uniform pointwise Lipschitz loss. This joint score condition is stronger than closeness of unconditional losses at a fixed decision, which would not control a report-dependent readout. These are consistent raw coupling conditions, not a Lipschitz bound on posterior normalization.

Theorem 4.2 (Discount–expansion modulus). *Under (16) and the stated raw report and loss bounds, a constant C independent of M and of the Borel controller satisfies*

$$(17) \quad |J_{\theta, \beta}(\gamma) - J_{\theta', \beta}(\gamma)| \leq C \begin{cases} s, & \beta L < 1, \\ s(1 + |\log s|), & \beta L = 1, \\ s^{\log(1/\beta)/\log L}, & \beta L > 1, \end{cases} \quad 0 < s \leq \frac{1}{2}.$$

The same bounds compare robust values on parameter sets at Hausdorff distance at most s in such a common family. The three orders are sharp in the class of bounded expanding raw experiments. The constants need not be uniform as $\beta \uparrow 1$.

Proof. Use the same action and label uniforms until the first report disagreement, and maximal report couplings at each call. While the reports agree, all retained states, actions and readouts agree, even when their rows are discontinuous. The probability of a disagreement by t is at most $C' L^t s$, since the geometric sum of previous raw report errors is of this order. Outside the disagreement event the loss difference has the same bound. Hence the stage risk difference is at most $\min(1, C'' L^t s)$.

Sum this bound against $(1 - \beta)\beta^{t-1}$. Split at the integer T for which $L^T s$ first becomes of order one. The early sum is bounded by a constant times $s \sum_{t \leq T} (\beta L)^{t-1}$; the late sum is at most a constant times β^T . A geometric sum, or T terms in the critical case, gives (17). Hausdorff comparison follows by comparing each fixed controller on the two sets, then taking infima without changing that controller class.

For sharpness, take $X_0 = s \in [0, 1]$, $X_{t+1} = \min(1, L X_t)$, reports independent uniform noise, a singleton action and decision, and stage loss X_t . The comparison with preparation 0 is exactly $(1 - \beta) \sum_{t \geq 1} \beta^{t-1} \min(1, L^t s)$. Its early terms and tail have the three matching orders just computed. This is a raw-modulus sharpness witness, not a claim that its memory problem is difficult. $\square$

For example, controlled logistic maps $x \mapsto \theta_a x(1 - x)$ on $[0, 1]$, with $0 \leq \theta_a \leq 4$, satisfy $|X_t^\theta - X_t^{\theta'}| \leq s(4^t - 1)/12$ from a common preparation. Reports $Y_t = X_t + U_t$, where U_t is uniform on $[0, w]$ and $w > 0$, have total variation at most $|X_t - X_t'|/w$. Squared prediction loss on $[0, 1]$ is 2-Lipschitz in the hidden target. Thus Theorems 2.2 and 4.2 apply with $L = 4$, even in expanding regimes and although report supports move. For a finite prefix the explicit bound

$$\Omega_N(s) = (1 - \beta) \sum_{t=1}^N \beta^{t-1} \min \left\{ 1, \sum_{r=1}^t \frac{s(4^r - 1)}{12w} + \frac{s(4^t - 1)}{6} \right\}$$

is controller-uniform. Uniform report-bin approximation on a compact enclosing interval has one-call error at most $2h/w$, from the two boundary cells. Coupling it over t calls gives $\min(1, 2th/w)$, an alternative to the tensor bound (11). With computable preparation integrals these are effective certificates. No convergence rate for learning θ is inferred.

4.2. Noncommuting instruments with changing rank.

Theorem 4.3 (Projective and positive quantum observations). *On a finite-dimensional Hilbert space let a prepared density matrix evolve through trace-preserving quantum instruments with densities*

$$\mathcal{I}_{\theta,t}^a(y)(\rho) = \sum_{r=1}^R K_{\theta,t,a,r}(y) \rho K_{\theta,t,a,r}(y)^*, \quad \int \sum_r K_{\theta,t,a,r}(y)^* K_{\theta,t,a,r}(y) d\sigma_a(y) = I.$$

No strictly positive lower effect bound is assumed. Let the declared score expectation at a decision be a bounded continuous-in-decision effect evaluated on the post-measurement state. This is one executable terminal audit at the selected scoring call; the unscored trajectory is not subjected to unrecorded earlier audits. Then Theorem 2.2 applies to the classical M -state observer. Uniform computable instrument approximations and model moduli imply Theorem 3.3, including projective effects and ranks changing with state, report or parameter.

Proof. For a fixed action word, composition of the unnormalized completely positive maps gives a positive trace-class matrix density on the product report space. Its trace is the report density and integrates to one. Applying the bounded scoring effect gives the payoff density. Continuity in the decision follows before any division by a report probability. Thus zero-probability events and rank changes cause no denominator problem. Each payoff submeasure is defined by its own terminal audit after that report prefix. A joint classical path of outcomes for incompatible quantum audits is neither used nor assumed. If an audit is actually performed before later calls and changes the state, it must instead be included in the intervening physical instrument.

For the effective assertion, replace maps before normalization. If $\|K_r(y)\|, \|K'_r(y)\| \leq B$, then, for a positive state ρ of trace one,

$$\|K_r \rho K_r^* - K'_r \rho K_r'^*\|_1 \leq 2B \|K_r - K'_r\|.$$

Summing and integrating gives a raw instrument error bound. Positive trace-preserving instruments, followed by any stochastic controller rows, contract the direct-sum trace norm on Hermitian inputs. Telescoping gives the sum of the one-call errors for every fixed controller. Common parameter and report approximations therefore provide the required moduli, irrespective of posterior rank or contraction. This argument uses neither an invented hidden classical quantum trajectory nor a positive inverse effect. $\square$

A concrete natural instance has two noncommuting measurement actions, with projectors P_θ^a and $I - P_\theta^a$, and report maps $\rho \mapsto P_\theta^a \rho P_\theta^a$ and its complement. The common reference assigns mass $1/2$ to each of the two reports (so the instrument densities are twice the displayed maps) and dominates all reports, but an outcome can be impossible for some states and parameters.

For rotating projectors, $\|P_\theta^a - P_{\theta'}^a\| \leq C|\theta - \theta'|$, so the sum of the two branch errors is at most $4C|\theta - \theta'|$. Report discretization error is zero. Pure preparations and repeated unitary evolution are permitted. This example fails a uniform $\kappa I \leq E_y$ hypothesis, yet satisfies the present theorem. The physical density matrix belongs to the experiment; M is classical controller memory, not unrestricted quantum memory.

Corollary 4.4 (Singular reference and flagged strata). *The response theorem remains valid for a common Cantor report reference, a countable report reference, or a finite mixture of mutually singular references with observed stratum tags, provided finite histories have the corresponding product densities. Vanishing stratum weights and rank changes are allowed. Effective conditional-expectation certificates use the actual cylinder or tail masses.*

Proof. The tensor and averaging proofs use only a finite product of probability spaces and L^1 approximation; neither has a Lebesgue-density step. Tagged finite mixtures are probability references on a disjoint union. Zero mixture weights simply set some response densities to zero. For countable reports, retain the tail as a report cell and use its supplied uniform actual mass to bound coarsening; do not condition it away. The quantum construction above supplies genuine changing-rank branches. These verify the hypotheses directly. $\square$

5. ACTUALLY ACQUIRED GEOMETRY AND A NONDOMINATED SINGULAR CLASS

The raw-response value is a task invariant, not automatically a dimension formula. Matching numerical rates require additional mass and implementation structure. We give a completely proved singular case and retain the broader finite-horizon acquired-mass and stability results in the companion [5] with their original hypotheses.

Let S be a compact subset of $\mathbb{R}^d$ and let D be a compact convex set containing S . The prepared target X has law μ_θ on S . The first paid acquisition reports (**prepare**, X). Every subsequent paid call reports only **blank**. At every time the task is squared prediction of the same X . Rescale the diameter, if necessary, so the loss is at most one. The visible preparation tag is part of the stated physical experiment, not a secretly supplied clock. The controller is exactly the stationary machine of Definition 2.1.

Theorem 5.1 (Robust blind acquired geometry). *For every arbitrary family $(\mu_\theta)_{\theta \in \Theta}$ on S , every $M \geq 1$, and every $0 < \beta < 1$,*

$$(18) \quad V_\beta^\Theta(M) = \min_{c_1, \dots, c_M \in D} \sup_{\theta \in \Theta} \int \min_{1 \leq j \leq M} \|x - c_j\|^2 d\mu_\theta(x).$$

One parameter-independent nearest-center encoder followed by self-loops attains the value. The same formula is the optimal checkpoint risk immediately after the first acquisition. No domination of the preparation family is required. In particular, it includes the uncountable nondominated family $\mu_\theta = \delta_\theta$ on a continuum, when the right side is the squared covering radius of that actual family of preparations.

Proof. For any Borel controller, its output at every time belongs to its fixed set of at most M readouts. Conditional on $X = x$, even a randomized internal trajectory incurs expected loss at least $\min_j \|x - d_j\|^2$. Integrate this pointwise inequality, sum the discount weights, and take the parameter supremum. This proves the lower bound for every controller, including randomized reuse of its states.

For any centers, choose the least-index nearest center at the preparation report. The encoder is Borel by continuity of the distances. On blank reports every state loops to itself. Its risk equals the displayed integral at every time and every parameter. The initial label can be one of these same M labels: the preparation tag, not an extra register, distinguishes the first update. The checkpoint has the same lower and upper construction. Finally, for each parameter the displayed integral is

continuous on compact D^M by bounded convergence; its supremum is lower semicontinuous and attains a minimum. This proof is direct and does not invoke product domination from Theorem 2.2 in the nondominated case. $\square$

Corollary 5.2 (Mass, covering and singular memory rates). *For a fixed actual acquired law μ , suppose its support has a global covering bound $N(r) \leq C_1 r^{-s}$ at small r , and it has a submeasure $\lambda \leq \mu$ of mass $m > 0$ such that $\lambda(B(x, r)) \leq C_2 r^s$ for every center and sufficiently small r . Then for all sufficiently large M ,*

$$(19) \quad \frac{m}{2} \left(\frac{m}{2C_2 M} \right)^{2/s} \leq V_{\beta}^{\{\mu\}}(M) = q_M(\mu) \leq C_3 M^{-2/s},$$

with constants independent of β . The exponent may be nonintegral. The factor involving m is retained; a rare stratum is not normalized into a full-probability experiment.

Proof. Choose $r = (m/(2C_2 M))^{1/s}$. The union of M balls has λ -mass at most $m/2$, so at least $m/2$ mass has squared error at least r^2 , regardless of randomization. For the upper, choose a covering with at most M centers at radius bounded by a constant times $M^{-1/s}$ and use the encoder from Theorem 5.1. These are actual-law lower and globally implementable upper bounds in the same task. $\square$

For the uniform middle-thirds Cantor preparation, a depth- k cylinder has mass 2^{-k} and diameter 3^{-k} . These cylinders give $N(r) \leq C r^{-\log 2 / \log 3}$. A ball with radius between $3^{-(k+1)}$ and 3^{-k} intersects a uniformly bounded number of depth- k cylinders, giving the matching small-ball bound. Hence (19) applies with $s = \log 2 / \log 3$, without a smooth carrier. Products of such preparations and ordinary bounded-density coordinates give anisotropic cylinder coverings; (18) retains their entire quantization curve rather than replacing it by ambient dimension. Atomic mixtures and accumulating atoms are included directly by (18); a power law is not asserted when the mass and covering hypotheses fail.

For the general continuing experiment, a singleton Bayes task also gives a useful lower diagnostic. Let $m_t = \mathbb{E}[Z_t \mid H_t]$ for a bounded Euclidean target, under a fixed declared acquisition policy independent of compression. Conditional expectation gives

$$\mathbb{E}\|Z_t - d_{I_t}\|^2 = \mathbb{E}\|Z_t - m_t\|^2 + \mathbb{E}\|m_t - d_{I_t}\|^2 \geq B_t + q_M(\text{Law}(m_t)),$$

provided the full history includes the controller's used randomization, which is independent of the hidden target conditional on its physical reports and actions. A discounted sum preserves this inequality. For adaptive optimization, the acquired law must be recomputed under each deployed policy; this fixed-policy inequality is not an unrestricted control theorem. Nor may one subtract separately optimized Bayes baselines in the robust problem. The exact blind theorem avoids both issues by proving its matching construction and lower bound in the very same experiment.

6. CAUSAL SIMULATION, NUMERICAL ERROR AND PHYSICAL ACQUISITIONS

Define $\text{TV}(P, Q) = \sup_B |P(B) - Q(B)|$. A simulator is parameter independent, nonanticipative, and supplied with an action lift, report transformation, internal state, task readout and physical clock certificate. The comparison below includes the scored outcome, not just the marginal report law. At call t a scored prefix means the actions and reports through t , the declared readout, and the one task audit executed there. It does not require a joint law of unperformed, possibly incompatible earlier audits. The run used to execute this audit then ends; the underlying continuing controller itself has no hidden audit clock.

Theorem 6.1 (Discount-weighted resource transfer). *Suppose a source can simulate the first t calls of a target for every common target controller, with total-variation error at most ε_t in the corresponding scored prefix, at most S persistent simulator states, and at most $C(t)$ source*

acquisitions. Assume the simulator can run continuously with these same state and action conventions. Then for the pulled-back target-call score,

$$(20) \quad V_{\text{source},\beta}^{\Theta}(MS; C) \leq V_{\text{target},\beta}^{\Theta}(M) + (1 - \beta) \sum_{t \geq 1} \beta^{t-1} \min(1, \varepsilon_t).$$

The notation on the left specifies source acquisitions indexed by virtual target calls. It is not a claim that source physical-time discount equals target-call discount. A reverse lower comparison requires a reverse typed simulator.

If the t th conditional simulation error is at most e_t , an admissible prefix allowance is

$$\min\{1, \sum_{r \leq t} e_r\}.$$

Uniform error e therefore contributes at most $\min(1, e/(1 - \beta))$. Composing simulators multiplies their persistent-state bounds, composes acquisition bounds, and adds their certified prefix deficiencies before discount weighting.

Proof. The source executes the product of the target state and the simulator state; the action lift and report transformation use neither θ nor future reports. The target readout is issued at the certified completion of each virtual call. For a score in $[0, 1]$, the difference of expected scores at call t is at most the scored-prefix total variation. Discount summation gives (20) for each fixed target controller and model. Take the model supremum and then an infimum over target controllers; no minimax exchange is used. Coupling until the first conditional mismatch gives the prefix sum. The identity $(1 - \beta) \sum_{t \geq 1} t\beta^{t-1} = 1/(1 - \beta)$ gives the uniform bound. The serial simulator has the product state; the triangle inequality and contraction under postprocessing add deficiencies. Clock and acquisition claims follow only from the supplied completion certificates. $\square$

Exactly the same proof treats numerical report classification. If, under the ideal deployed law, the probability of the first classification mismatch at call r is at most χ_r , its discounted risk allowance is at most $(1 - \beta) \sum_{r \leq t} \beta^{r-1} \min(1, \sum_{r \leq t} \chi_r)$. Uniform output loss error adds its loss modulus. A model misspecification of radius δ adds the appropriate controller-uniform modulus, such as (17), not automatically δ^2 . In the exact-deadline setting a small error alone would not preserve legality; this is why the interface boundary is explicit here.

Proposition 6.2 (Sharp discounted amplification by actual erasure). *At each call let a fresh fair bit be the scored target. The ideal experiment reports it. In a second experiment an irreversible hidden failure occurs independently with probability e at each still-operating call, before the report. Before failure the bit is reported; from failure onward the report is an erasure symbol. Scores are not feedback. For squared prediction, a three-state continuing controller attains*

$$(21) \quad V_{\beta,\text{erase}}(3) = \frac{e}{4(1 - \beta + \beta e)}, \quad V_{\beta,\text{ideal}}(2) = 0.$$

The scored-prefix total variation under the identity report comparison is exactly $1 - (1 - e)^t$. The displayed risk is a lower bound even for an unrestricted-history observer, hence remains exact for any larger finite budget.

Proof. The first erasure event is visible. Its probability by call t is $1 - (1 - e)^t$. On its complement the full scored prefix is the ideal one, and on the event its report prefix is disjoint from the ideal support; this proves the exact total variation. Conditional on erasure the fresh bit remains fair and independent of all observations, giving unavoidable squared risk $1/4$. Otherwise prediction is exact. States with readouts $0, 1, 1/2$, overwritten by each report, attain these conditional optima. Summing $[1 - (1 - e)^t]/4$ with the discount weights gives (21). The one-step raw kernels can be defined to agree on the already-failed state and differ by e on an operating state, so the

conditional perturbation bound is genuine. This is an attained channel error, not an allowed tolerance promoted to a lower bound. $\square$

There is an operational interpretation of the discount which also fixes acquisition accounting. An independent external selector τ with $\mathbb{P}(\tau = t) = (1 - \beta)\beta^{t-1}$ gives $J_{\theta, \beta} = \mathbb{E}\ell(d_{I_\tau}, Z_\tau)$ and $\mathbb{E}\tau = 1/(1 - \beta)$ actual calls. The selector is external experimental machinery; its randomness and timing are not a free internal controller. Under a pathwise simulator certificate the expected source calls are at most $(1 - \beta)\sum_t \beta^{t-1}C(t)$. A deterministic truncation at N has omitted probability β^N , but it is a different stopping interface. The prefix bound in Theorem 2.2 truncates the score in analysis and leaves the implementing table unchanged. An infinite continuing execution has infinitely many raw acquisitions and physical calls; it is not described as a hard- N experiment. Finite physical-time conclusions require the stated per-call or $C(t)$ time bounds.

7. A COMPLETE CONTINUOUS-REPORT CERTIFICATE

This example makes the global lower, model uncertainty and implementing upper fully explicit. It is a calibration and quantization witness, not a substitute for the general theorem or a claim to solve a difficult nonlinear filter.

At every call a fresh fair hidden bit X_t is prepared. For a fixed inaccessible $p \in [p_0, p_1] \subset (0, 1)$ the report density on $[0, 1]$ is

$$(22) \quad g_p(y \mid x) = 1 + (2x - 1)p(2y - 1).$$

There is one action. Scores $(X_t - d)^2$ are not additional reports. The report sequence is iid uniform for every p , so its past cannot reveal the current bit or the parameter. With the known endpoint p_0 , not the realized p , an equally spaced M -cell partition has conditional posterior means

$$d_j = \frac{1}{2} + \frac{p_0}{2} \left(-1 + \frac{2j - 1}{M} \right), \quad 1 \leq j \leq M.$$

Every controller row simply stores the current cell. All M states are reused; no count is retained.

Proposition 7.1 (Exact continuous acquired-memory law). *For decision space $[0, 1]$, every $M \geq 1$ and $0 < \beta < 1$,*

$$(23) \quad V_{\beta}^{[p_0, p_1]}(M) = \frac{1}{4} - \frac{p_0^2}{12}(1 - M^{-2}).$$

The common unrestricted-history value is $B^ = 1/4 - p_0^2/12$, so the memory excess is exactly $p_0^2/(12M^2)$. An inaccessible fixed calibration offset $u \in [-\delta, \delta]$ in the scored target $X_t + u$, not in its reports, adds exactly δ^2 to the robust value when decisions are allowed in $[-\delta, 1 + \delta]$.*

Proof. At $p = p_0$, the posterior mean is $1/2 + p_0(2Y_t - 1)/2$, uniform on an interval of length p_0 , and all earlier reports and controller randomization are independent of the current target/report pair. Its irreducible Bayes risk is $1/4 - p_0^2/12$. An M -valued output has an additional quantization error at least $p_0^2/(12M^2)$: cells of lengths l_j have minimal uniform second moments $l_j^3/12$, and $\sum l_j^3 \geq p_0^3/M^2$. Randomized cells cannot improve nearest-center assignment. This is a lower bound for every state-reusing controller at every call.

For the displayed equal cells, write $m(Y)$ for the midpoint of the cell in the variable $2Y - 1$. Then $\mathbb{E}m^2 = \mathbb{E}[m(2Y - 1)] = (1 - M^{-2})/3$. The risk of the common readout is

$$\frac{1}{4} + \frac{p_0^2}{4}\mathbb{E}m^2 - \frac{pp_0}{2}\mathbb{E}m^2,$$

which is largest at the known endpoint p_0 and attains (23). Allowing full real-valued reports corresponds to $m = 2Y - 1$ and proves the stated common baseline, rather than assuming a modelwise oracle baseline is attainable.

For the offset lower, average the risks at $u = \delta$ and $u = -\delta$ with $p = p_0$. Their mean is the unshifted risk plus δ^2 , for any common controller. The unshifted optimum remains the same even with the larger readout interval, since projection onto $[0, 1]$ decreases its loss. The displayed common forecast has zero mean error at every p , so its risk at fixed u is its unshifted risk plus u^2 . This gives the matching upper. The shifted squared loss is bounded by $(1 + 2\delta)^2$; scale it to use the unit-loss normalization of the general theorem. $\square$

7.1. Finite certificate with rational arithmetic. Take $\beta = 1/2$, $M = 2$, $p \in [1/4, 1/2]$, and the declared finite decision space $D = \{7/16, 1/2, 9/16\}$. This finite space has the same optimum as (23) for $M = 2$. Use the two report cells $[0, 1/2)$ and $[1/2, 1]$, denominator $Q = 2$, and the two endpoint models. There are exactly

$$|D|^2 \binom{2+2-1}{2-1}^{2 \cdot 2} = 9 \cdot 81 = 729$$

listed tables, with initial label 1. Every probability and discounted risk in this list is rational. For a table with next-label probabilities k_{icj} , its label transition matrix is $T_{ij} = (k_{i0j} + k_{i1j})/2$; its one-call score vector $r_p(i)$ is obtained by integrating (22) over each half interval with readout d_j . The exact discounted score is

$$(24) \quad J_{p,1/2}(g) = \frac{1}{2} e_1^\top (I - T/2)^{-1} r_p.$$

The inverse exists because T is stochastic. Since T is independent of p and r_p is affine in p , the maximum over the whole interval is exactly at an endpoint. This is an exact model reduction, so its parameter allowance is $\Omega = 0$, not a sampled approximation assumption.

Exhaustive rational evaluation of all 729 tables gives $L = U = 63/256$, with evaluation error $\zeta = 0$. One selected table has readouts $(7/16, 9/16)$, and for both previous labels sets $k_{i0} = (1, 0)$ and $k_{i1} = (0, 1)$. The table is parameter independent and uses no random bits at execution.

For clarity, the generic uniform report error and an instance-specific value reduction are different. Here the payoff density is affine in its last report, with slope $(p/2)(1 - 2d)$ in the variable $2y - 1$. For $d \in D$, its two-cell L^1 projection error is at most $1/128$. The bound (11) therefore gives $E_\infty \leq 1/64$. The generic rounding estimate at $Q = 2$ is too coarse to be useful. Instead, an exact value-preserving rounding certificate is available: for any fixed pair of readouts, iid reports make old labels irrelevant to the current conditional loss, and each cell's expected loss is linear in its next-label probabilities. At p_0 a minimizing cell rule is deterministic; the symmetric pair and sign rule minimize there and have their worst risk there. Thus the robust cell optimum is attained among the listed deterministic rows, so the *value* rounding allowance is zero. This does not say that every Borel row is uniformly close to a deterministic row.

Using this proved refinement in the same lower-comparison argument yields the certified continuous-family interval

$$[L - E_\infty, U] = [59/256, 63/256].$$

The independent analytic all-Borel lower in Proposition 7.1 tightens it to $[63/256, 63/256]$. No temporal truncation error is used in this example because (24) evaluates the infinite series exactly. Its expected acquisitions under geometric scoring are 2; this is not a deterministic two-call budget. The stored forecasts $7/16$ and $9/16$ use the known family endpoint p_0 . Both the finite enumeration and its exact rational output are deposited separately from the continuum proof.

8. COMPARISON AND MATHEMATICAL BOUNDARIES

The static-reduction method of Witsenhausen [6] moves a dynamic observation law into a cost density under an independent reference. Strategic-measure compactness and existence in decentralized control were developed in [9, 8]; approximation by independently dominated

measurements is studied in [3]. These are predecessors of the response expansion and tensor argument here. We do not claim the independent-reference reduction, weak-star compactness, or the finite-intersection principle as new. The point requiring verification for the present question is that a single finite row is reused at all times and models, while the physical state is not assumed finite-dimensional. Proposition 2.3 proves this verification before discounting; Theorem 3.3 keeps the original state budget on both sides of a global approximation.

Robust finite-state controllers for uncertain finite POMDPs are studied in [2], and inductive finite-controller synthesis in [1]. Those works address controller design and useful algorithms; the present exhaustive certificate is not a replacement for their computational methods. Finite-window near optimality in [4] compares increasing windows with an unrestricted POMDP value under filter stability. Here the comparison is against all Borel controllers at a fixed cardinality, without a filter-stability hypothesis, but with product-history domination and supplied effective response data. Average-cost near optimality in [7] has a genuinely different objective and assumptions; no limit $\beta \uparrow 1$ in our theorem supplies its conclusion.

Theory	Declared comparison	Distinction relevant here
Static reduction and existence [6, 8]	Sequential or decentralized policies under reference-measure hypotheses	Foundational method; not itself the stated robust fixed-reused-row certificate
Robust FSC synthesis [2]	One finite controller for uncertain finite POMDPs	Robust control is established; our report space can be continuous and our hidden presentation infinite-dimensional
Inductive synthesis [1]	Finite-state POMDP controllers and indefinite-horizon specifications	Algorithmic synthesis; our lower comparison covers all Borel rows at unchanged M
Finite-window control [4]	Window policies versus unrestricted history, with stability	Different memory class and baseline; we do not infer a learning or window-convergence rate
Present exact/effective pair	One stationary M -state controller; fixed arbitrary model family; strictly discounted continuation	Attained robust value, finite-model obstruction; effective same- M lower and upper under extra computable data

The parameter is fixed along a path in every statement about our robust value. A timewise adversary, an internally timed exact deadline, and an unrestricted quantum-memory controller are different comparison classes. Exact Borel controllers may have no finite description. Our effective layer supplies finite tables, but table parsing, numerical report classification and integration have separate cost certificates. The numerical example is evidence that the two endpoints can be instantiated, not a proof of the general theorem.

The following boundaries remain mathematical questions rather than consequences proved here. The exact theorem does not cover arbitrary nondominated report families; the nondominated statement in Theorem 5.1 has its own direct proof. We do not have a general undiscounted average-cost or internally stopped theorem, a uniform $\beta \uparrow 1$ compactness conclusion for optimal values, a classification by an intrinsic numerical dimension for all raw experiments, unknown-kernel learning rates, or a matched optimum for description length, workspace and physical time. Nor is a witness-size bound extracted from compactness or a practical complexity guarantee from enumeration. The discount–expansion estimate gives a quantitative consequence for unstable hidden dynamics, not a resolution of every robust filtering problem. The finite-horizon geometric and causal-deficiency results in the complete companion retain their own hypotheses and full proofs. These distinctions isolate the central advance without replacing the original experimental question by a particular realization.

REFERENCES

- [1] R. Andriushchenko, M. Češka, S. Junges, and J.-P. Katoen, *Inductive synthesis of finite-state controllers for POMDPs*, Proceedings of Machine Learning Research **180** (2022), 85–95.
- [2] M. Cubuktepe, N. Jansen, S. Junges, A. Marandi, M. Suilen, and U. Topcu, *Robust finite-state controllers for uncertain POMDPs*, Proceedings of the AAAI Conference on Artificial Intelligence **35** (2021), 11792–11800, doi:10.1609/aaai.v35i13.17401.
- [3] I. Hogeboom-Burr and S. Yüksel, *Sequential stochastic control (single or multi-agent) problems nearly admit change of measures with independent measurement*, Applied Mathematics & Optimization **87** (2023), article 51, doi:10.1007/s00245-023-09965-5.
- [4] A. D. Kara and S. Yüksel, *Near optimality of finite memory feedback policies in partially observed Markov decision processes*, Journal of Machine Learning Research **23** (2022), no. 11, 1–46.
- [5] Q. Qi, *General Theta Foundations I: Acquired geometry and causal resource transfer. Finite-horizon common-controller allocation and effective completion*, complete companion manuscript, 2026. The separately preserved finite-horizon article contains the exact allocation, compact predictive realization, regular and singular acquired-geometry, and typed simulation proofs.
- [6] H. S. Witsenhausen, *Equivalent stochastic control problems*, Mathematics of Control, Signals and Systems **1** (1988), 3–11, doi:10.1007/BF02551232.
- [7] H. Yu and D. P. Bertsekas, *On near optimality of the set of finite-state controllers for average cost POMDP*, Mathematics of Operations Research **33** (2008), 1–11, doi:10.1287/moor.1070.0279.
- [8] S. Yüksel, *A universal dynamic program and refined existence results for decentralized stochastic control*, SIAM Journal on Control and Optimization **58** (2020), 2711–2739, doi:10.1137/18M1221382.
- [9] S. Yüksel and N. Saldi, *Convex analysis in decentralized stochastic control, strategic measures, and optimal solutions*, SIAM Journal on Control and Optimization **55** (2017), 1–28, doi:10.1137/15M1049129.